Alina Lynn Kochocki

Curriculum Vitae (Fall 2026)

CONTACT INFORMATION

Institution: Loyola Marymount University
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EMPLOYMENT

Loyola Marymount University (LMU) *August 2026 – May 2027*
Adjunct Lecturer, Department of Physics

EDUCATION

University of California, Los Angeles (UCLA) *September 2026 – Present*
Enrolled in graduate real analysis via UCLA Extension

Michigan State University (MSU) *September 2021 – August 2026*
Advised by Nathan Whitehorn
Doctor of Philosophy in Physics. GPA: 3.89
Dissertation focusing on modeling and statistical inference for partially observed systems, domain applications

Technical University of Munich (TUM) *June 2025 – June 2026*
Visiting graduate student within the group of Elisa Resconi.
Audited graduate coursework in Nonlinear Dynamics, Dynamical Systems, and Continuous-Time Markov Processes

University of California, Los Angeles *September 2017 – March 2021*
Advised by Rene Ong, Nathan Whitehorn and Alexander Kusenko
Bachelor of Science in Physics. GPA: 3.89
Magna Cum Laude, Highest Departmental Honors

RESEARCH INTERESTS

- Branching and tree-valued processes
- Coalescent and genealogical structures
- Interacting particle systems
- Adaptive network structures, transport, and interaction on networks
- Scaling limits and mean-field approximations
- Statistical inference for complex and partially observed systems

SELECTED PREPRINTS, DOCTORAL WORK

- *(Under review at the Journal of Theoretical Biology)* Alina Kochocki. 2026 “Threshold Resource Redistribution in Spatially-Structured Kinship Networks”.
- *(In prep)* Alina Kochocki. 2026 “Statistical Methods in High-Energy Physics”. IceCube technical note.

OTHER SELECTED PUBLICATIONS

- *(Under review at PRD)* Alina Kochocki and Xavier Rodrigues. 2026 “An Evolving Leptonic Jet Model for Delayed Radio Flares in Neutrino Blazars”.

- *(Collaboration review)* The IceCube Collaboration. 2026 "A Time-Dependent Search for Neutrino Emission from Flaring X-ray Binaries".[1].
- *(Under review at ApJ)* Alina Kochocki, Sam Hori, Emma Kun. 2026 "Characterizing Gamma-Radio Delayed Flaring Activity from Blazars".
- The IceCube Collaboration and Atacama Cosmology Telescope. 2026 "A Search for Millimeter-Bright Blazars as Astrophysical Neutrino Sources". *ApJ*, 999, p.98
- Alina Kochocki et al. for the IceCube Collaboration. 2025 "A Search for Astrophysical Neutrinos from Flaring X-ray Binaries with IceCube". *PoS*, ICRC, 1077
- Alina Kochocki for the IceCube-Gen2 Collaboration. 2023 "Forecasted Sensitivity of IceCube-Gen2 to the Astrophysical Diffuse Spectrum". *PoS*, ECRS, 100
- Alina Kochocki, Volodymyr Takhistov, Alexander Kusenko and Nathan Whitehorn. 2021 "Contribution of Secondary Neutrinos from Line-of-Sight Cosmic Ray Interactions to the IceCube Diffuse Astrophysical Flux". *ApJ*, 914, p.91

GRADUATE RESEARCH

- **Branching Populations**. My most recent work focuses on particle branching processes with interactions. I am especially interested in tree- or genealogy-dependent interactions, in which population structure influences its evolution. I am investigating this mechanism both statistically and from a probabilistic perspective.
- **Characterization of Jetted, Astrophysical Sources**. I developed likelihood-based analyses to search for time-dependent, high-energy neutrino emission from jetted, active galactic nuclei (AGN) and X-ray binaries. I performed new analyses of Fermi-LAT gamma-ray data for a large population of AGN, and studied time-series correlations between gamma-ray and radio data, providing some evidence for delayed radio activity. I constructed a dynamic, multi-zone jet model to describe evolving leptonic spectra and radiative emission from selected sources.
- **Event Reconstruction with Generative Models**. Observable event interactions within the IceCube detector are typically reconstructed as compositions of stochastic radiative losses and Cherenkov emission. Photon propagation exhibits location- and direction-dependent scattering and absorption coefficients. I developed a generative network model for radiative predictions of point interactions. This work included Monte Carlo simulation, model development and training, implementation within our likelihood-based reconstruction algorithm, and performance comparison with existing, table-based methods.
- **Study of Diffuse Cosmic-Ray and Astrophysical Fluxes**. I initially explored potential, large-scale instrumentation upgrades to the IceCube Neutrino Observatory. I characterized the performance of these potential, expanded configurations through forecasts of sensitivity to the diffuse, astrophysical neutrino flux. I then became interested in atmospheric neutrinos, a substantial background to astrophysical signal. I developed new simulations of cosmic-ray air showers containing a rare prompt neutrino interaction. I studied the observability and distinguishability of this unique signature – containing both shower and neutrino interaction. I explored potential analyses incorporating this topological information.
- **Optical Testing for the IceCube Upgrade**. I oversaw commissioning of the IceCube Upgrade mDOM optical module (MSU production lot) optical and environmental testing – 2M USD in sensor hardware. I organized and mentored a group of undergraduate students to support testing. I led the first studies of sensor timing resolution.

[1]IceCube authorship policies are described here. The listed papers represent first-author contributions.

AWARDS AND HONORS

NTTF Professional Development (Delphi) Award, Loyola Marymount University — October 2026
Dissertation Completion Fellowship, Michigan State University — October 2025
Michigan State University STEM Ambassador, STEMAP — February 2023
National Science Foundation GRFP – Honorable Mention — April 2022
University Distinguished Fellowship, Michigan State University — February 2021
College of Natural Science Fellowship, Michigan State University — February 2021
Recruiting Fellowship, Michigan State University — February 2021
Selected Graduate Fellowship Offers (declined) — February 2021

- **James Mills Peirce Fellowship**, Harvard University
- **Stanford Graduate Fellowship**, Stanford University
- **KIPAC Graduate Fellowship**, Stanford University
- **Princeton Graduate Fellowship**, Princeton University

USRA Distinguished Undergraduate Award – Honorable Mention — August 2020
Undergraduate Research Scholar, UCLA URC — August 2020
Honors Summer Research Stipend Award, UCLA College Honors — June 2020
Undergraduate Research Scholar, UCLA URC — August 2019
Frederick R. Waingrow Peterson Publishing Company Scholarship — April 2019
Undergraduate Research Fellow, UCLA URC — December 2018
Invited Member of UCLA College Honors Program — April 2017

SELECTED TALKS

- AMS Fall Western Sectional Meeting (Tempe, USA) – October 2025.
- *Three plenaries and sixteen parallel presentations between 2022 and 2025 at IceCube collaboration meetings. Meetings are typically attended by ∼200 collaborators and are held twice yearly – within and outside of the United States.*
- International Cosmic Ray Conference (Geneva, Switzerland) – July 2025.
- (***Invited***) Northwestern CIERA Theory Seminar (Evanston, USA) – April 2025.
- (***Invited***) UCLA TEPAPP Seminar (Los Angeles, USA) – March 2025.
- American Physical Society Global Physics Summit (Anaheim, USA) – March 2025.
- (***Invited***) PACIFIC (Moorea, French Polynesia) – August 2024.
- (***Invited***) Turbulence in Astrophysical Environments (Santa Barbara, USA) – February 2024.
- TeV Particle Astrophysics (Napoli, Italy) – September 2023.
- 27th European Cosmic Ray Symposium (Nijmegen, the Netherlands) – July 2022.
- 237th Meeting of the American Astronomical Society – January 2021.
- UCLA Undergraduate Research Virtual Showcase – May 2020.

TEACHING EXPERIENCE AND MENTORSHIP

Instructor of Record (LMU) — *August 2026 – May 2027*

- *Spring 2027:* PHYS 1100: Introduction to Mechanics (Lower-division, 4 units) – Two sections
- *Fall 2026:* PHYS 3100: Electrodynamics (Upper-division, 4 units)
- *Fall 2026:* PHYS 2100: Introduction to Electricity and Magnetism (Lower-division, 4 units, includes laboratory)

Undergraduate Research Mentorship (MSU) — *May 2023 – September 2025*
Closely advised Amiri Walker and Dhiti Doddamreddy on IceCube data analysis projects. Trained and advised a group of eight undergraduate students to prepare and perform laboratory data acquisition.

IceCube Summer School (University of Wisconsin–Madison) *June 2024*
Prepared and presented lectures on "Topics in the Neutrino-Sources Working Group" and "Point-Source Analyses and Likelihood".

Elementary Particle Physics Laboratory (UCLA) *September 2020 – December 2020*
Contributed to development of the virtual course syllabus structure and content. Responsible for creation and oversight of the simulation component of this course; produced a cookbook for the construction of optical physics simulations catered towards students with minimal background in experimental particle physics. Produced original Python analysis and Geant4 simulation tutorials presented in class.

RECENT STUDIES IN MATHEMATICS

Real Analysis – UCLA;
Probability, Continuous-Time Markov Processes – TUM; Audited and ongoing self-study
Linear Algebra – UCLA
Nonlinear Dynamics, Dynamical Systems – TUM; Audited
Network Analysis – Independent study (online lecture series)
Modeling Environmental and Social Systems – MSU

RECENT STUDIES IN PHYSICS

Statistical Mechanics – UCLA; ongoing review
Quantum Field Theory – MSU
Classical Mechanics – MSU
Radiative Astrophysics – MSU
Extragalactic Astrophysics and Cosmology – MSU
ISAPP 2023: Neutrino Physics, Astrophysics and Cosmology – INFN School, Varenna
Special Topics in High-Energy Physics, Collider Phenomenology – MSU

TECHNICAL SKILLS

Mathematical Methods: Monte Carlo methods, likelihood-based inference, stochastic simulation, numerical simulation, numerical optimization
Programming Languages: C, C++, Python, Bash, Perl, Java, Boost (Python bindings via Boost.Python)
Mathematical Computing Software: Mathematica, SOLIDWORKS
Relevant Libraries/Frameworks: TensorFlow, XGBoost, BLAS, LAPACK, Celerite2, NumPy, SciPy
Scientific Software: CORSIKA, Pythia, Geant4, Arduino, SCPI
Domain-Specific Data Processing: ROOT (particle physics data processing), IceTray (high-energy neutrino and cosmic-ray data processing), VEGAS and Fermitools (gamma-ray data processing)
Distributed Computing: TORQUE, Sun Grid Engine, Slurm, HTCondor
Computing Experience: Extensive experience with CPU/GPU simulations and model training. Efficient parallelization of tasks across resources.
Communication: GitHub, LaTeX, Overleaf, Wiki Markup, Microsoft Office, Google Workspace
Backend/Frontend Development: HTML, CSS

SERVICE

IceCube Neutrino-Sources Technical Lead *March 2024 – February 2026*
Technical lead for the IceCube Collaboration Neutrino-Sources (astrophysics) working group. Oversaw software reviews for data analyses, provided assistance with software, likelihood implementation, data and computing resources.

MSU Physics Graduate Organization President *July 2024 – July 2025*
President of the MSU Physics Graduate Organization. Involvement in departmental advisory. Oversight of graduate student participation in departmental faculty hiring. Supported student-organized services and representatives. Assisted departmental leadership in communicating effectively with the student

body during periods of funding instability and policy changes. Provided students with resources during periods of instability. Led survey to understand student needs and experiences.

Graduate Colloquium Committee Representative *July 2023 – October 2024*
Organized and hosted student-postdoctoral lunches with visiting colloquium speakers. Worked with local colloquium committee to improve departmental colloquium experience and attendance.

HEP Social Organization *July 2023 – December 2024*
Hosted socials and happy hours for the high-energy physics (HEP) group at MSU.

OUTREACH

Astronomy on Tap, "IceCube and Astrophysical Neutrinos" *July 2024*
Invited to speak on IceCube and neutrino astrophysics for a public audience in East Lansing, Michigan.

Outreach Activity, "Academic Research and Community with IceCube" *January 2023*
Developed an outreach event for high school students at St. Catherine of Siena Academy (Wixom, MI). Presented on my own work with IceCube, entering science and academia, and led a guided discussion with small groups to consider the lifestyle of a researcher, bridging potential misconceptions held about STEM education and careers. Assessed activity efficacy with an IRB-approved survey.

STEM Ambassador Program *September 2022 – February 2023*
Accepted into the 2022 cohort of the NSF-supported STEM Ambassador Program (STEMAP). Attended a set of regular workshops to learn best practices for outreach and for building community trust in science. Developed and carried out a novel outreach activity.

UCLA Exploring Your Universe (EYU) Science Fair *November 2018, 2020*
Presenter at UCLA's Exploring Your Universe fair. Helped manage organization of the 2020 cosmology and astroparticle physics virtual booth.

Head Telescope Operator, UCLA Public Viewing Nights *September 2017 – February 2019*
Organized weekly shows, presented scientific background to audiences, and mentored club members.

LANGUAGES

English – Native
Spanish – Limited working proficiency (five years of classroom study)
German – Elementary proficiency (B1 level)

HOBBIES

Running and weight training for dance; sketching, water and oil painting.